

SIMONE MARIA GIANCOLA

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CURRENT

PhD researcher (doctorant)

Orsay institute of mathematics (IMO)

Nov 2025 - present

Orsay, FRA

- Advisors: C. Giraud, N. Verzelen (INRAE Montpellier)
- Statistical to computational gaps, low-degree method, statistical physics

EDUCATION

Orsay institute of mathematics (IMO), Univ. Paris-Saclay

M2 probability and statistics

2024 – 2025

Orsay, FRA

Bocconi University

MS data science

2021 – 2024

Milan, ITA

Arizona State University

Undegraduate exchange

2021

Phoenix, USA

Bocconi University

BS economics, management and computer science

2018 - 2021

Milan, ITA

EXPERIENCE

Research intern

with C. Giraud, N. Verzelen (INRAE Montpellier), A. Carpentier (Potsdam)

Apr 2025 - July 2025

IMO Orsay, FRA

Research intern

with P. Richtárik

Feb 2024 - May 2024

KAUST, KSA

Research intern

with B. Loureiro

Oct 2023 - Dec 2023

ENS Paris Ulm, FRA

Research intern

with M. Mondelli

Jun 2023 - Aug 2023

IST Austria, AUT

PAPERS

Carpentier, **Giancola**, Giraud, and Verzelen. *Low-degree lower bounds via almost orthonormal bases*. [arXiv:2509.09353 \[stat.ML\]](https://arxiv.org/abs/2509.09353), 2025.

Richtárik, **Giancola**, Lubczyk, and Yadav. *Local curvature descent: squeezing more curvature out of standard and Polyak gradient descent*. In: **NeurIPS25, preprint**, 2024.

TEACHING

Analysis, (x2) Teaching assistant, R209

IUT d'Orsay, 2026

Graph theory, (x2) Teaching assistant, R207

IUT d'Orsay, 2026

DISTINCTIONS

Selected for the 12th Heidelberg laureate forum (HLF).

Scholarships: OeAD Austria ref. num. MPC-2023-01128; first Italian school in geometric deep learning.

TALKS

Séminaire des doctorants, Low-degree lower bounds via almost orthonormal bases *Paris Cité*, 2026

Group seminar, Low-degree lower bounds via almost orthonormal bases *Potsdam*, 2025

Stochastic gradient descent methods, Local curvature descent *KAUST*, 2024

Séminaire des doctorants, IT and algorithmic limits of mixed linear regression *ENS Paris*, 2023

Group seminar, IT and algorithmic limits of mixed linear regression *IST Austria*, 2023

SERVICE

Reviewer NeurIPS 2024, ICLR 2025.

Mentor Prison math project (PMP); LeadTheFuture.

SKILLS

Advanced Python, $\text{\LaTeX}$;

Basic Julia, Git, Unix, R, SQL, Matlab, C++, Keras, TensorFlow.

LANGUAGES

English (proficient); French (intermediate); Spanish (basic); Italian (native).

PROFILE & INTERESTS

Motivated researcher with a background in statistics, computer science, and probability.

Passionate about the interplay of rigorous research and insights from physics. In my spare time, I enjoy rugby, motorbike trips, podcasts, reading, and running.